NMC-OPT STUDIO: SCIENTIFIC NOVELTY & ARCHITECTURAL INNOVATIONS

Academic Research Contributions in Data-Movement-Aware Compiler Optimization

Publication Category: Academic Research Contributions & Innovation Audit
Core Scope: Compiler Design, Intermediate Representation, and Near-Memory Cost Modeling
Repository: https://github.com/MSN-1307/NMC-OPT-CD | **Target:** Heterogeneous CPU/PIM Systems

1. EXECUTIVE STATEMENT OF NOVELTY

While Processing-In-Memory (PIM) and Near-Memory Computing (NMC) hardware prototypes (such as UPMEM DPU, Samsung HBM-PIM, and SK Hynix AiM) have demonstrated extraordinary bandwidth gains in hardware laboratories, software toolchains remain in their infancy. Existing compilers (e.g., GCC, Clang/LLVM) treat memory accesses as uniform-cost load/store operations, lacking the analytical machinery required to make autonomous offloading decisions between host CPUs and distributed in-memory compute units.

NMC-Opt Studio bridges this fundamental software-hardware gap by introducing an adaptive, data-movement-aware compiler optimization workbench. The platform is not an ad-hoc hardware simulator, but an explicit compiler analytical framework that formalizes interconnect transit, bank interleaving, and layout reorganization costs directly inside the compiler optimization decision pass.

2. CORE RESEARCH & TECHNICAL INNOVATIONS

- **Data-Movement-Aware Target Selection Matrix:** Traditional compilers optimize strictly for operation latency and register allocation. NMC-Opt Studio elevates interconnect data traffic (MB transferred) and operational intensity (FLOP/Byte) to first-class compiler objective functions.
- **Unified Heterogeneous SSA Intermediate Representation (IR):** The framework introduces a specialized 3-address SSA IR that captures memory semantics across both host memory and distributed PIM banks with explicit barrier synchronization instructions.
- **First-Principles Analytical Cost Engine with Interconnect Tax:** Replaces slow cycle-accurate emulators with a closed-form first-principles analytical model factoring in host DMA dispatch overheads, physical link bandwidth constraints, and barrier latency.

2. CORE RESEARCH & TECHNICAL INNOVATIONS (CONTINUED)

- **In-Memory Layout Reorganization & Bank Striping Model:** Explicitly models the compile-time cost of converting contiguous host memory buffers into 16-bank interleaved layouts (T_{layout}), preventing invalid offload decisions that assume zero-cost data rearrangements.
- **Context-Aware Compiler Copilot & Educational Workbench:** Integrates an interactive compiler decision walkthrough with a retrieval-augmented Compiler Copilot grounding real-time parameter adjustments in compiler design principles and peer-reviewed literature.

3. STATE-OF-THE-ART COMPARATIVE MATRIX

Capability / Dimension	GCC / Clang	UPMEM SDK	NMC-Opt Studio
Target Selection	Host CPU only	Manual pragma	**Autonomous Analytical Matrix**
Data Movement Model	Zero link cost	Manual host-device DMA	**Explicit First-Principles Tax**
Bank Interleaving	Ignored	Manual bank scatter	**Compile-Time Layout Model**
Compilation Latency	Milliseconds	Seconds	**Real-Time (<50 ms)**
Intermediate Representation	Standard SSA	Clang LLVM IR	**Bank-Aware 3-Address SSA**

4. SUMMARY OF ACADEMIC & CURRICULAR IMPACT

NMC-Opt Studio establishes that future compiler architectures must evolve beyond pure instruction scheduling to incorporate interconnect and memory topology awareness. By providing an open, fully inspectable, and reproducible software workbench, the project enables both compiler researchers and students to explore heterogeneous memory-centric computing with mathematical rigor.